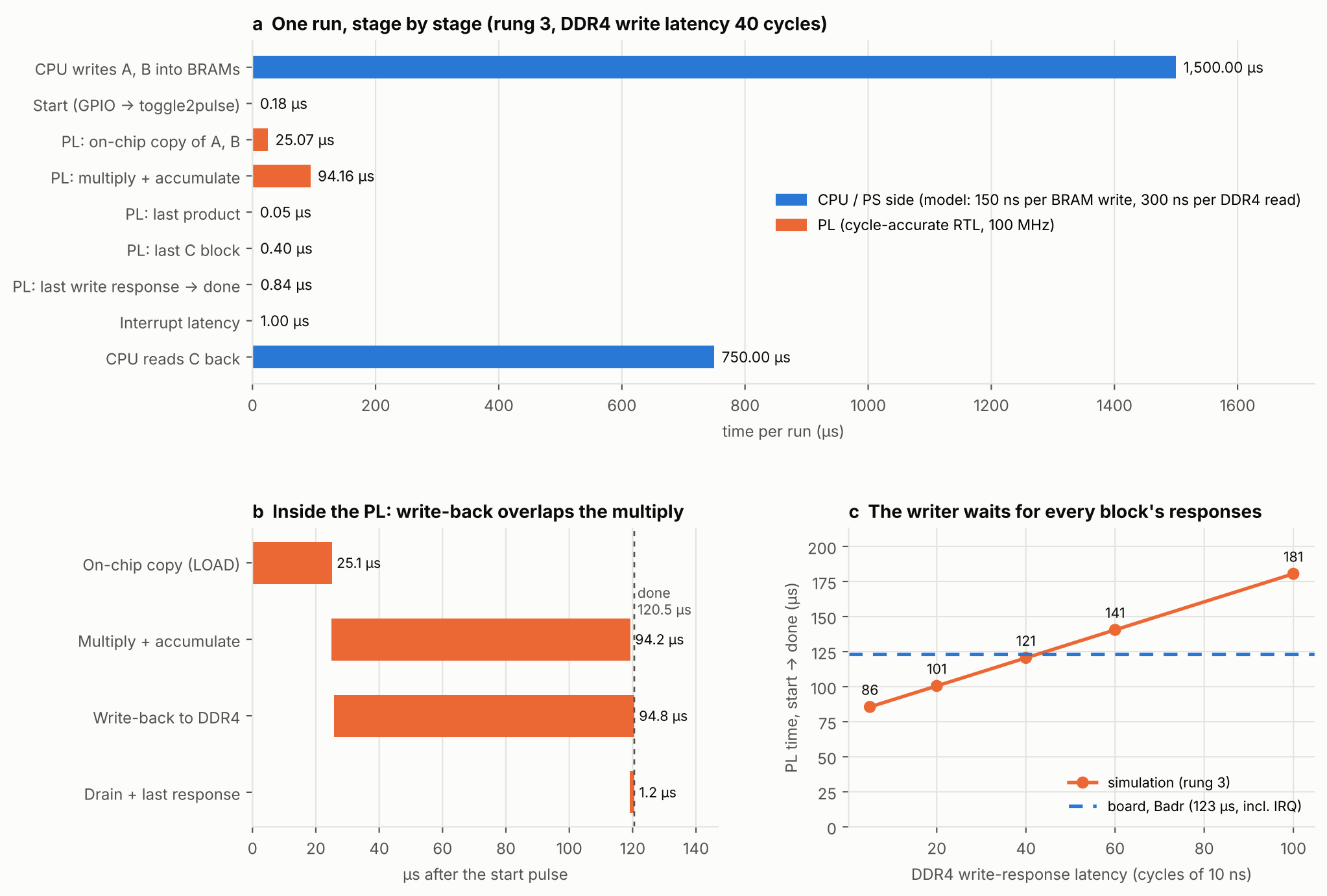
a One run, stage by stage (rung 3, DDR4 write latency 40 cycles)
CPU writes A, B into BRAMs
Start (GPIO → toggle2pulse)
PL: on-chip copy of A, B
PL: multiply + accumulate
PL: last product
PL: last C block
PL: last write response → done
Interrupt latency
CPU reads C back
1,500.00 µs
0.18 µs
25.07 µs
94.16 µs
0.05 µs
0.40 µs
0.84 µs
1.00 µs
750.00 µs
CPU / PS side (model: 150 ns per BRAM write, 300 ns per DDR4 read)
PL (cycle-accurate RTL, 100 MHz)
0
200
400
600
800
1000
1200
1400
1600
time per run (µs)
b Inside the PL: write-back overlaps the multiply
On-chip copy (LOAD)
Multiply + accumulate
Write-back to DDR4
Drain + last response
25.1 µs
done
120.5 µs
94.2 µs
94.8 µs
1.2 µs
0
20
40
60
80
100
120
140
µs after the start pulse
c The writer waits for every block's responses
PL time, start → done (µs)
200
175
150
125
100
75
50
25
0
86
101
121
141
181
simulation (rung 3)
board, Badr (123 µs, incl. IRQ)
20
40
60
80
100
DDR4 write-response latency (cycles of 10 ns)